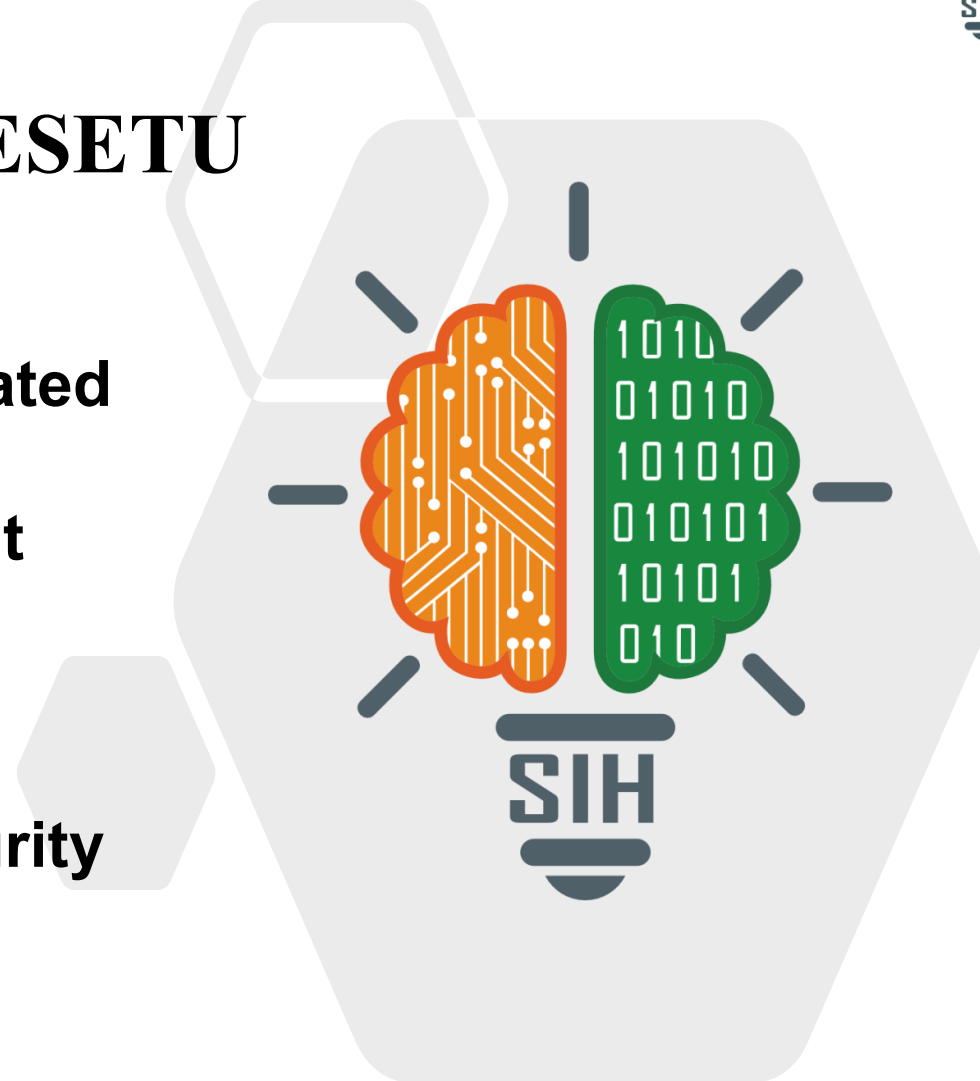


SMART INDIA HACKATHON 2026



TRACESETU

- **Problem Statement ID – PS26182**
- **Problem Statement Title – Automated Attribution of Unknown Cryptocurrency Wallets to Nearest Virtual Asset Service Providers (VASPs) through Blockchain Intelligence APIs**
- **Theme – Blockchain & Cybersecurity**
- **PS Category – Software**
- **Team ID – 170835**
- **Team Name – ANANTHA**



TRACESETU

The missing link is who controls the receiving address

The officer starts here

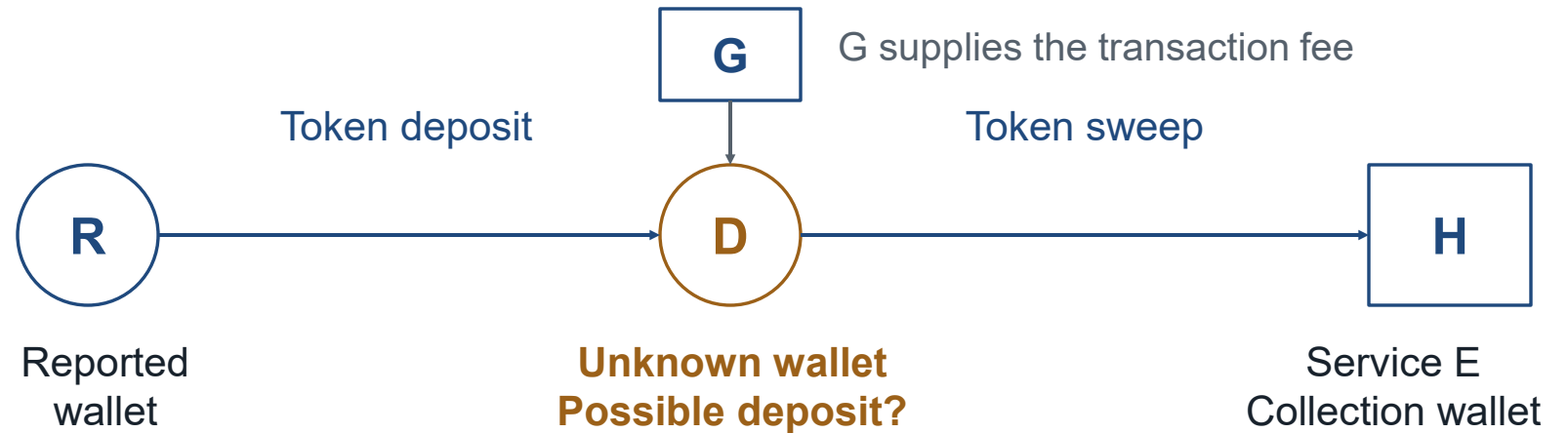
A complaint names a wallet.
The ledger shows its transfers.
Ownership can remain unknown.

The required answer

Which exchange or custodian received the funds, at which address, and what evidence supports that conclusion?

VASP means a company that provides services such as crypto exchange or custody.

Worked example: all addresses and service names are fictional



Where “Service E” comes from

A sourced register links G and H to E, with their roles and dates. Public tags or reviewed service confirmations supply the names. [4]

What TraceSetu investigates

Does D repeatedly receive tokens, get fees from G and sweep to H? If the checks pass, D becomes an unconfirmed deposit lead. [1, 2]

Seeded deposit-pattern matching + chronological graph search

1. Research test for deposit address D [1, 2]

Per cycle: G adds 100 native units before D sweeps

Cycle	Tokens in	Tokens to H	Fee paid
1	1,000	1,000	10
2	2,000	2,000	10
3	3,000	3,000	10

Token balance: $0 + 6,000 - 6,000 = 0$

Native balance: $0 + 300 - 30 = 270$

Require complete attested history, actual fee receipts, matching balances and dated G/H labels for service E. Reject conflicts, unexplained flows or missing evidence.

Synthetic integer units. Three cycles is an uncalibrated research setting, not a confidence percentage.

2. Find the first service on each path [3]

1 Start with R

Queue the address, asset, arrival event and path.

2 Expand one transfer at a time

Breadth-first search checks fewer hops first.

3 Keep ledger order

A later receipt cannot fund an earlier transfer.

4 Check the receiving address

Join dated service labels and deposit assessments.

5 Stop at custody or a gap

Return the service, address, path and unresolved leads.

“Nearest” means fewest observed transfers in scope.
A customer can imitate the pattern. D stays unconfirmed.

A 36-hour demo is feasible on the existing foundation

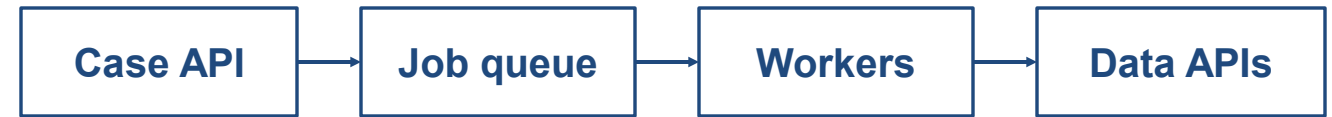
Hackathon scope

- 0–8 h** Acquire a bounded Bitcoin case
- 8–18 h** Trace paths and inspect labels
- 18–26 h** Challenge evidence and show gaps
- 26–32 h** Review and export a signed case
- 32–36 h** Test failures and rehearse

Research module: 64 synthetic checks pass.
Live integration and independent accuracy validation are still pending.

Public Bitcoin check: 2 transactions, 3 outputs.
Transfer data reproduced. No VASP established.
Six chain adapters exist, with unequal live coverage.

A path to Ministry deployment



Bound each search by time, hops and API budget. Cache replies.
Separate workers let case acquisition run outside the web request.
Shared quotas, agency fairness and load tests remain pending.

Tools and their jobs

Python / FastAPI	Evidence rules and authenticated APIs
React / TypeScript	UI and Cytoscape graph view
PostgreSQL + workers	Durable jobs; SQLite for the local demo
SHA-256 / Ed25519	Sign and verify evidence exports

National rollout also needs governed labels, SSO/MFA and approved SAHYOG access. Public/free APIs alone do not provide that coverage.

Our contribution: decide which evidence to check next

What changes the investigator’s answer

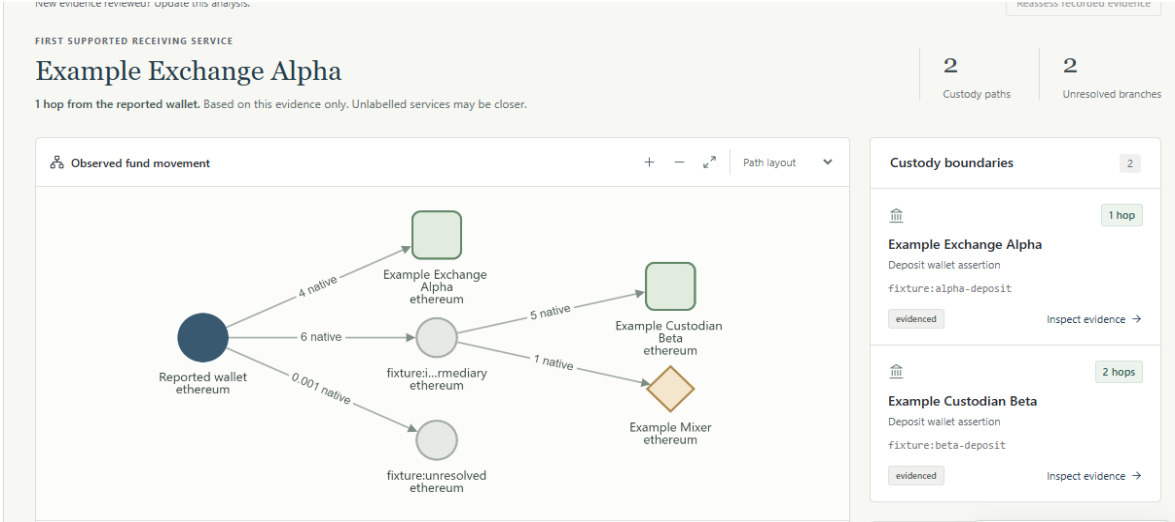
Evidence available	Result for the worked example
Known labels only	H: service label, 2 transfers away
Deposit pattern also passes	D: unconfirmed lead, 1 transfer
Remove G’s service evidence	D: unresolved; seek corroboration

The next-query planner prioritises nearer evidence gaps.
The source challenge reruns the case without a disputed label.
The certificate records what the search did and did not cover.

Testable benefit: preserve nearer deposit leads and expose the evidence that could change them.

Temporal graphs and clustering are established methods.
Our combined decision workflow still needs comparative evaluation.

The current investigation workspace



Actual app, fictional training case. The pattern detector remains a separate research module.

After review: a VASP request package

A verified recipient and independent reviewer precede export.
SAHYOG is not connected. Current exports remain NOT_SENT.

The method, its sources and the evidence still needed

[1] Victor, FC 2020

Deposit forwarding into known exchange addresses.

Motivates the unknown-deposit candidate test.

[2] Brechlin et al., 2025

Evonax fee funding and collection pattern.

The paper used service records to confirm address roles.

[3] Wu et al., PVLDB 2014

Time-respecting paths. Our bounded breadth-first search requires transfers to follow ledger order.

[4] GraphSense Public TagPacks

Public address labels with links to their original sources.

A proposed input for a reviewed service register.

[5] Lubbertsen et al., USENIX 2025

Evaluate ownership and deposit roles against independent ground truth, including coverage and missed addresses.

[6] Kappos et al., USENIX 2022

Bitcoin peel-chain clustering. A future guarded extension, not a capability of our current deposit-pattern module.

Working code and research

[github.com/
notvinayaks/TraceSetu](https://github.com/notvinayaks/TraceSetu)

Branch: product/live-foundation

Prototype video

[youtube.com/watch?
v=V_qB6dL934w](https://youtube.com/watch?v=V_qB6dL934w)

Next validation

Use confirmed deposits and look-alikes.
Measure correct service and role,
coverage, API calls and officer time.
No accuracy percentage is claimed.